

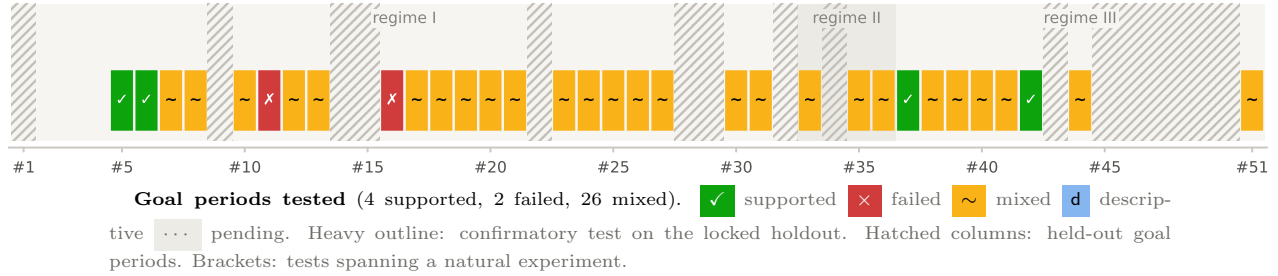
H34: Idea cascades follow a power law whose exponent reads off the distance to criticality

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 2/5 Usefulness 2.5/5

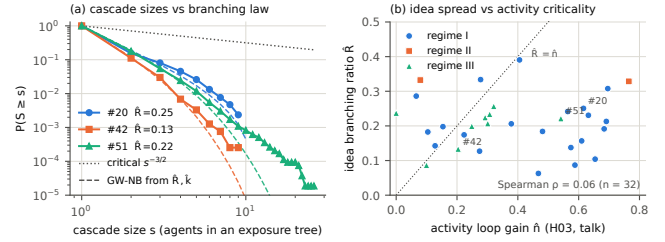
Bottom line. Idea cascades in the village are strongly subcritical ($\hat{R} \approx 0.2$), exposure-driven and heterogeneous rather than power-law. Branching ratio, not exponent or activity loop gain, is the number that forecasts how far an idea spreads.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

We hashed novel terms, URLs, numbers and names in 32 non-holdout goal periods (132k ideas). Each agent's first use was parented to the latest visible earlier use, giving exposure trees. Ideas spread subcritically everywhere: $\hat{R} = 0.06\text{--}0.39$, all CIs below 1. A new idea reaches a second agent 6–24% of the time. Adoption is time-locked to recent visible uses beyond a common field in 27/32 periods. Pure $s^{-3/2}$ is rejected in all 32; tails are heavier than one branching law. Failed: activity loop gains do not predict idea spread ($\rho = 0.06$); the pre-registered forecast was overconfident; simple vs complex contagion is unresolved. Holdout script written, not run.



Observables. (a) Fraction of exposure trees reaching at least s agents in #20, #42 and #51, with the branching law predicted from $\hat{R}, \hat{k}$ (dashed) and the critical $s^{-3/2}$ (dotted). Cascades are far from critical, and somewhat heavier-tailed than one branching law. (b) One dot per period. Idea branching $\hat{R}$ is unrelated to the activity loop gain $\hat{n}$ from H03; most periods lie below $\hat{R} = \hat{n}$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Contagion (model 03) as a Galton–Watson cascade (09) in a room of N agents. Adopters have negative-binomial offspring with mean R (branching ratio) and dispersion k . s : tree size. s_c : cutoff. h_k : adoption hazard per talk turn with k recent visible sources. R_c : contagion share of $\hat{R}$. η : daily drift of R in the forecast rule.

$$\hat{R} = \frac{\# \text{ first uses with an agent parent}}{\# \text{ agent first uses}}, \quad R_c = \hat{R} (1 - h_0/h_{\geq 1})$$

$$P(s | R, k) = \frac{\Gamma(ks + s - 1)}{\Gamma(ks)\Gamma(s + 1)} \frac{(R/k)^{s-1}}{(1 + R/k)^{ks+s-1}} \sim s^{-3/2} e^{-s/s_c}$$

$$s_c^{-1} = R - 1 - \ln R, \quad R_d = \hat{R} e^\eta, \quad \eta \sim \mathcal{N}(0, 0.25^2)$$

The usable output is a two-number forecast. Measure $\hat{R}$ (first uses with a visible parent, over all first uses) on the last two days of logs. At the village's $\hat{R} \approx 0.2$, expect 83% of new ideas or claims to stay with their author and about 1 in 150 to reach five agents. With day drift added, this post-hoc rule reaches 70–84% coverage at 90% nominal; it is frozen for the holdout. The negative lessons matter more. Activity criticality (Hawkes n) says nothing about idea spread. In a shared room, “who saw it first” cannot separate copying from independent invention without a timing test. A power-law exponent is just $\hat{R}$ in disguise. Nothing is near runaway: misinformation here dies locally unless it is an artifact everyone must touch.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 subcritical, $\hat{R} < 1$	$\hat{R}$ 0.06–0.39, CI < 1 in 32/32 periods, 113/113 class cells	pass
P1 order $U > N > D$	artifacts 0.35 > numbers 0.27 > names 0.19	fail
P2 $\hat{R} < \text{activity } \hat{n}$ (HH108)	25/32 periods	pass
P3a branching-law tail	$P(s \geq 3)$ in band 27/32; $P(s \geq 5)$ heavier in 25/32	fail
P3b no $s^{-3/2}$; $\tau_{\text{app}} \geq 2$	rejected 32/32; τ_{app} 2.4–4.3, $\rho(\tau_{\text{app}}, \hat{R}) = -0.95$	pass
P4 $\hat{R}$ tracks $\hat{n}$	$\rho = 0.06$ (Holm $p = 1$)	fail
P5 contagion beyond field	HR ₁₀ CI > 1 in 27/32 (regime I ≈ 2 ; II/III 14–75)	pass
P6 simple contagion	HR(2 vs 1) bracket 0.5–4.2	ambiguous
P7 forecast coverage $\geq 80\%$	36%/46%; post hoc V3 70%/84%	fail

GOAL PERIODS IN DETAIL

G05 ✓ supported
G06 ✓ supported
G07 ~ mixed
G08 ~ mixed
G10 ~ mixed
G11 × failed
G12 ~ mixed
G13 ~ mixed
G16 × failed
G17 ~ mixed
G18 ~ mixed
G19 ~ mixed
G20 ~ mixed
G21 ~ mixed
G23 ~ mixed
G24 ~ mixed

+16 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

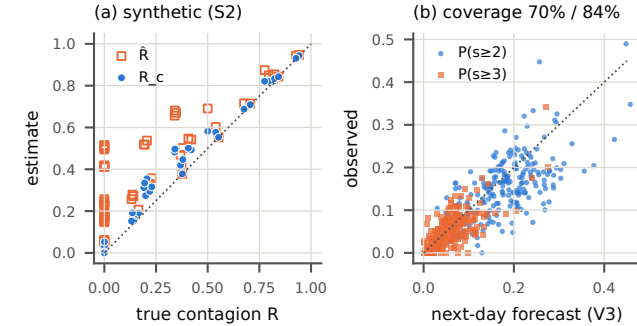
CAVEATS

“Ideas” are hashed marker rules, not semantic units, and $\hat{R}$ depends on the rule. Field vs contagion rests on HR₁₀: a bursty common trigger (a web page or human prompt reaching several agents at once) would mimic it, and copying from tools or the web is invisible. Visible exposure uses H18’s call-start rule, whose mention placebo failed. Novelty is relative to non-holdout text only. The H25 content-dial link (partial $\rho = 0.41$ given room size; daily $\rho = 0.28$) is post hoc and shares its source text with $\hat{R}$. Periods with rooms of 4 cannot show tails.

ROUND 2

Fit a mixture of idea-level R to explain the heavy tail. Predict $\hat{R}$ across the room split and merge before looking. Label numeric claims true or false to compare misinformation with its corrections.

- **Essence.** how far does a new idea or claim spread in an agent swarm, and what sets it: exposure, salience or a shared field?
- **H34-R1** Heterogeneous branching: fit a gamma mixture over idea-level R (one extra parameter) and test whether it removes the excess of trees with five or more agents and beats the beta-binomial on day-ahead log score.
- **H34-R2** Natural experiments for idea spread: the NE15 room split and the 05-04 merge, with the branching ratio and the per-pair branching per room member predicted from room size before looking (axis E).
- **H34-R3** Claims proper: label numeric claims as true or false (or corrected) with Jev and compare branching ratios of false claims and their corrections (the misinformation question).
- **H34-R4** Semantic ideas: embedding-cluster novelty, so paraphrased ideas count, with the same trees and HR10.



Second figure. (a) On real village timelines with synthetic ideas, a common field inflates $\hat{R}$ (squares) even with no contagion; R_c (dots) recovers the true contagion R . (b) The post-hoc rule’s next-day forecasts of $P(s \geq 2)$, $P(s \geq 3)$ vs observed, over 219 period-days.